

Unit II: Deep Learning for Computer Vision

Why Convolutions over FC Layers?

An intuitive exploration of the mathematical architecture
powering modern computer vision.

LEARNING OUTCOMES

By the end of this session, you will be able to:

- **Analyze** why Fully Connected (Dense) networks fail on high-resolution images.
- **Understand** the critical importance of 2D spatial relationships.
- **Explain** local connectivity and parameter sharing.
- **Grasp** the intuitive mechanics of the convolution (sliding window) operation.



LECTURE ROADMAP



SET INDUCTION

Consider this standard photograph...

It has a modest resolution of **1024 × 1024** pixels.

How many numerical values are stored in this image?

- Width: 1024
- Height: 1024
- Color Channels: 3 (Red, Green, Blue)

Total = 3,145,728 numbers



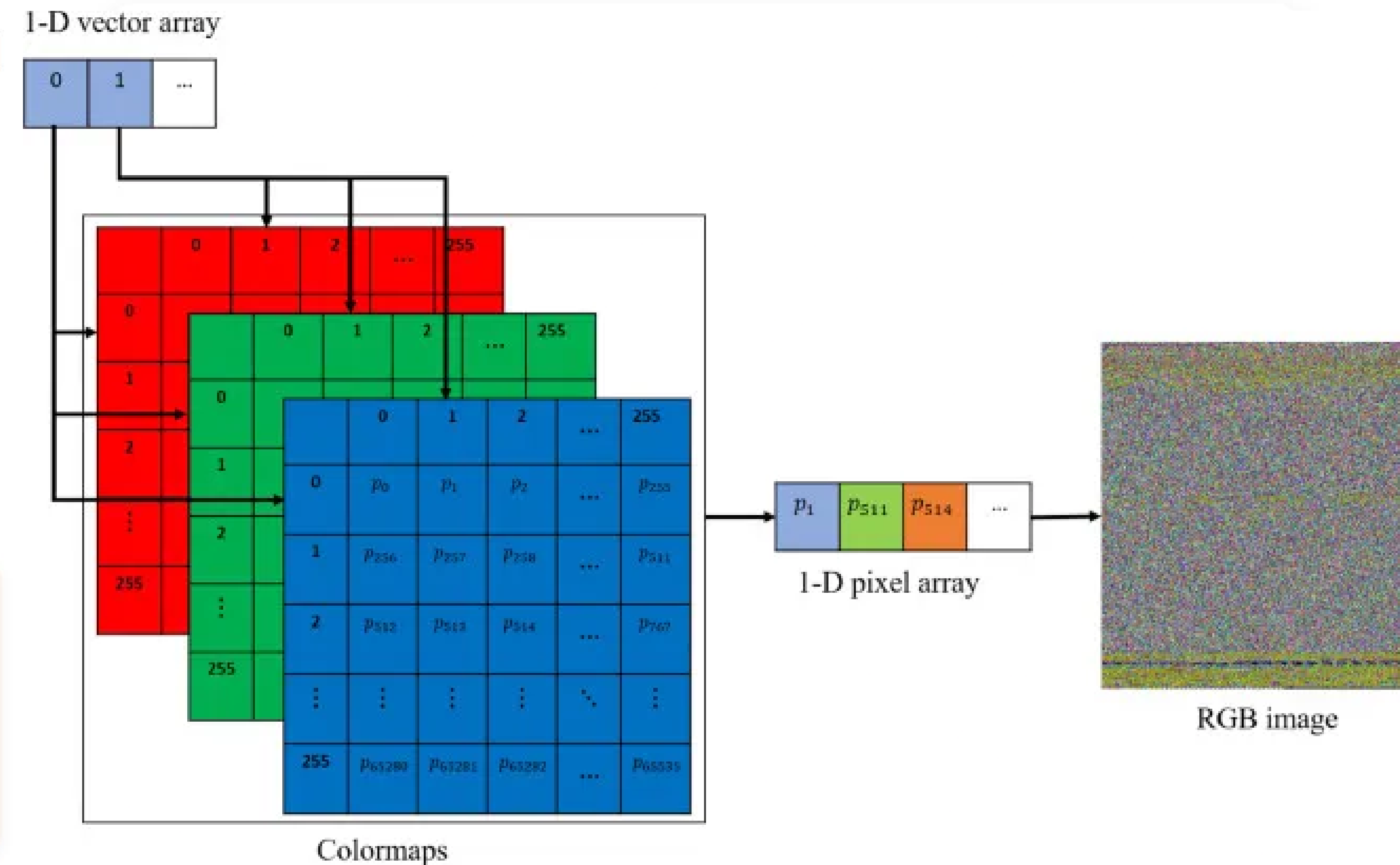
IMAGES AS NUMBERS

To a computer, an image is simply a multi-dimensional matrix (tensor) of intensity values.

- **Spatial Dimensions:** The 2D grid structure (height × width) dictates which pixels are neighbors.
- **Depth Dimension:** The color channels overlay perfectly aligned spatial data.

The Core Rule of Vision:

A pixel's meaning is entirely dependent on its immediate neighbors. A single dark pixel could be a shadow, an edge, or noise.



REVIEW: FULLY CONNECTED LAYERS

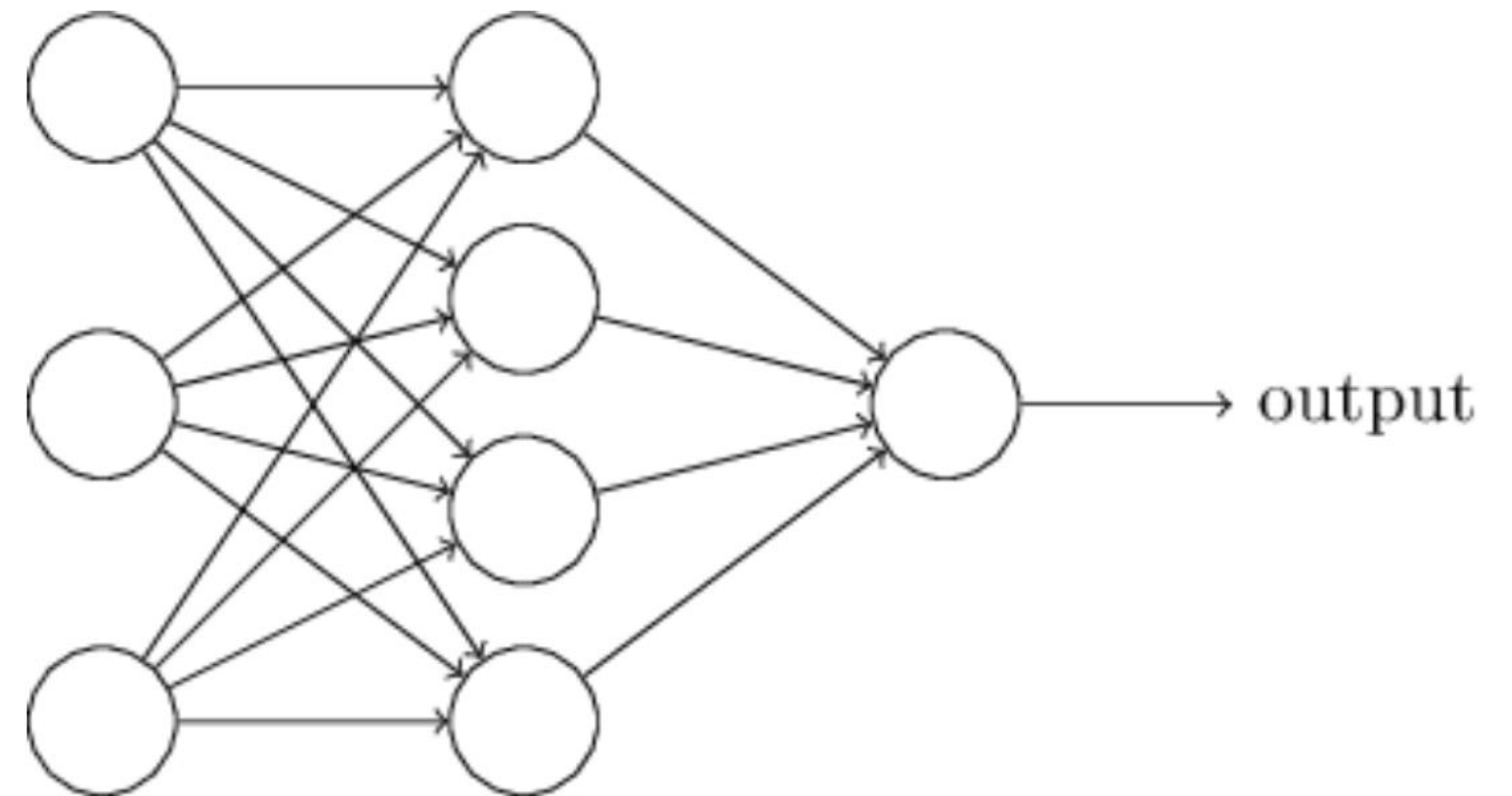
In standard Multilayer Perceptrons (MLPs), we use Dense or **Fully Connected (FC)** layers.

To feed an image into an FC layer, we must **flatten** it into a single 1D vector.

$[224 \times 224 \times 3] \rightarrow [150,528]$

The Fatal Flaw:

Flattening completely destroys all 2D spatial relationships. Pixels that were vertical neighbors are now placed far apart in the 1D array.



PROBLEMS WITH FC LAYERS



Parameter Explosion: Millions of required weights exceed hardware memory limits.



Severe Overfitting: Too many parameters cause the network to memorize training data rather than generalize.



Computational Bottleneck: Massive dense matrix multiplications make training agonizingly slow.



Loss of Spatial Info: As established, 1D vectors ignore the critical 2D structure of the physical world.

REAL-WORLD SCALING FAILURE

150M+

Weights for ONE Layer

The Mathematical Reality

Assume a tiny image size of **224×224 RGB** (standard for older models like VGG).

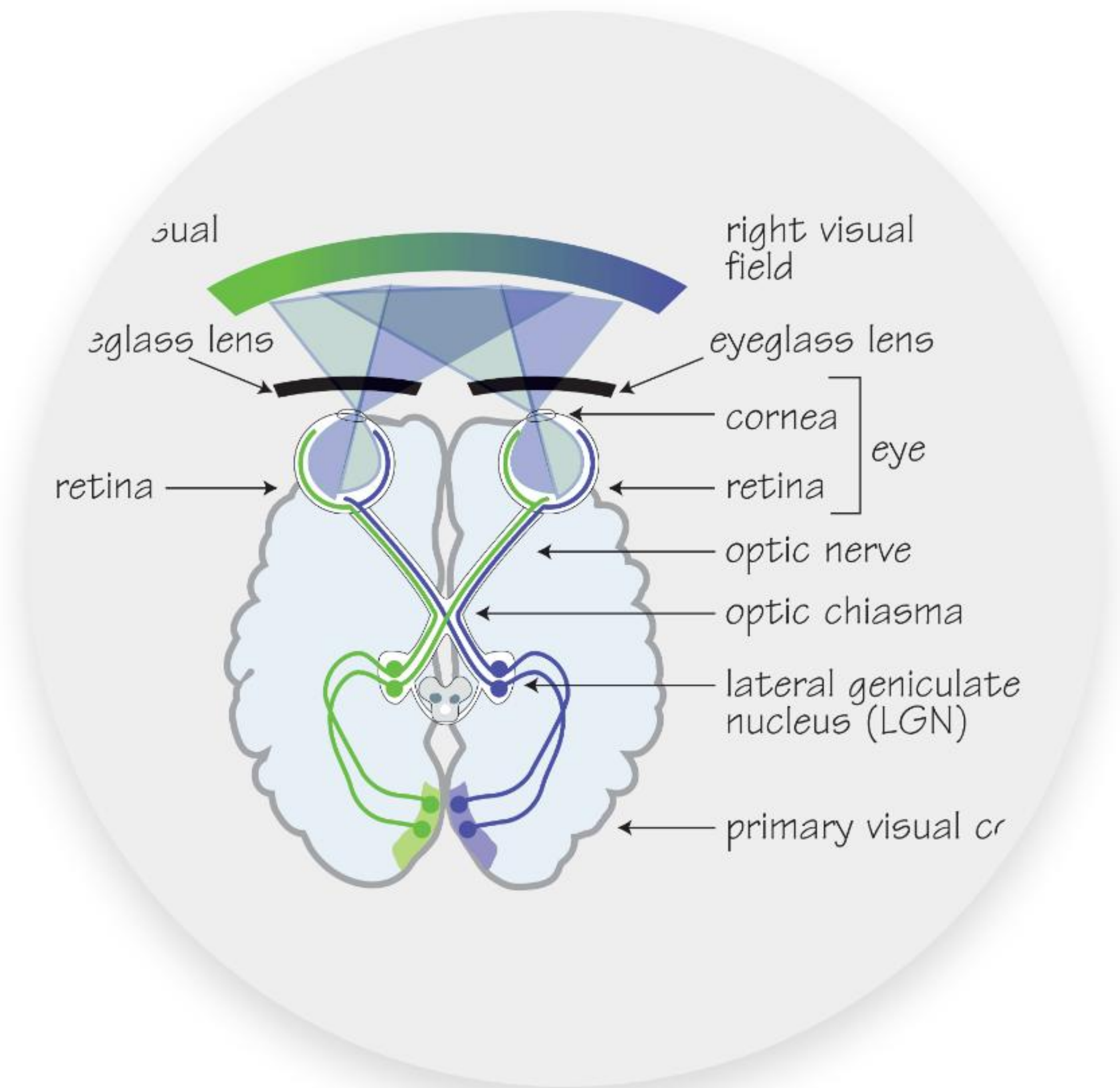
Inputs = $224 \times 224 \times 3 = 150,528$

If the first Hidden Layer has just **1,000 neurons** (which is very small):

- **150,528 inputs × 1,000 neurons**
- **= 150,528,000 parameters**

And that is just layer one.

THE BIOLOGICAL SOLUTION



Local Receptive Fields

Humans do not process every visual receptor simultaneously as a giant flat list.

Biological vision relies on neurons that only respond to a small, localized region of the visual field (a **receptive field**).

Did You Know?

In 1959, Hubel and Wiesel discovered that neurons in a cat's visual cortex only fired when exposed to specific localized edges and patterns, inspiring artificial convolutions.

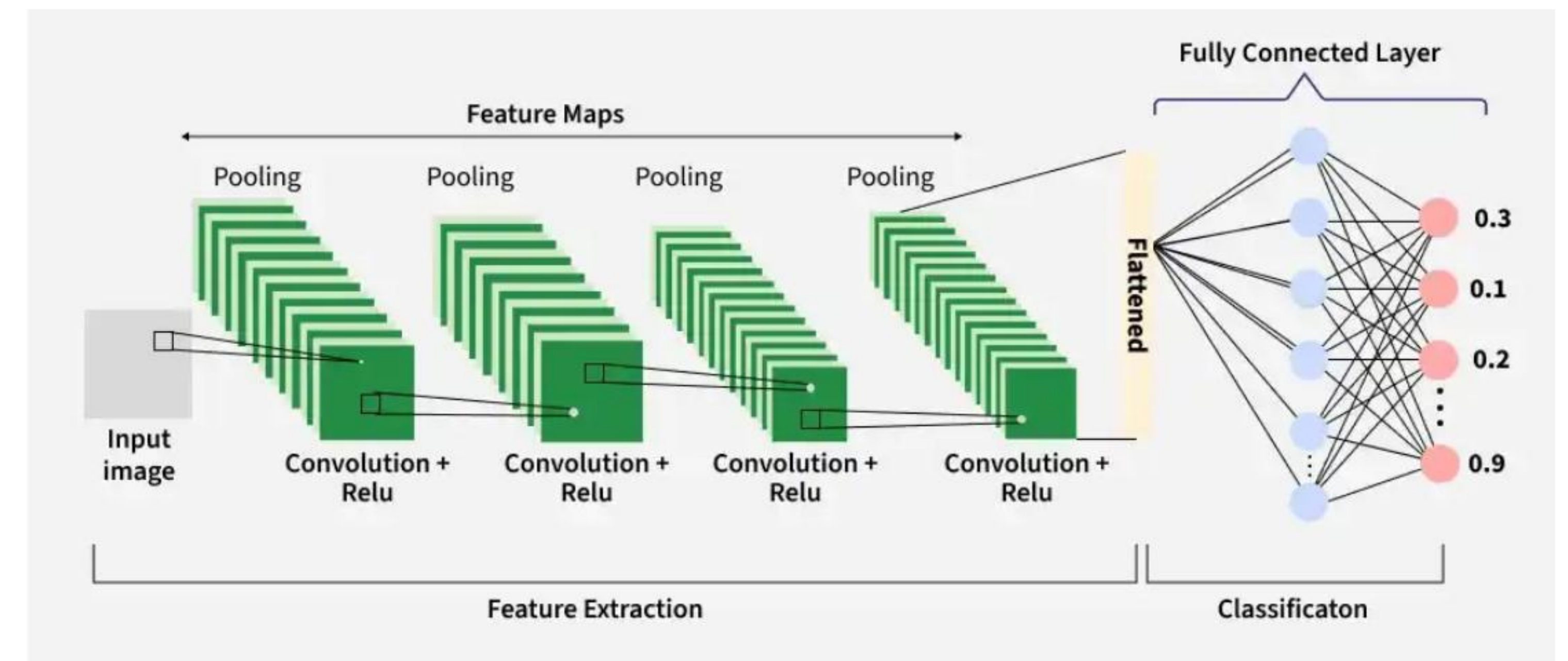
INTRODUCING CONVOLUTION

The Sliding Window

Instead of connecting a neuron to the entire image, we define a small **Kernel** (or filter), usually 3×3 or 5×5 pixels.

We **slide** this small kernel across the width and height of the image.

- It extracts local visual features.
- It completely preserves 2D structure.
- It solves the parameter explosion problem.



CORE CONCEPT 1: LOCAL CONNECTIVITY

Dense vs Local

In FC networks, a neuron connects to **every pixel**. It is overwhelmed by noise from unrelated sides of the image.

In CNNs, a neuron connects only to a **small neighborhood** (e.g., 3×3).

Why it works

In images, pixels are highly correlated with their immediate neighbors. A group of dark pixels forming a line is a meaningful feature (an edge).

We force the network to learn local patterns first.

Instead of finding "a cat in 150,000 pixels," we find "fur texture in a 3×3 patch."

CORE CONCEPT 2: PARAMETER SHARING

Without Sharing (FC)

If a dense network learns to detect a vertical edge in the top-left, it uses a specific set of weights.

If that same edge is in the bottom-right, it has to learn **different** weights.

With Sharing (CNN)

In a CNN, the exact same 3×3 Kernel slides over the **entire** image.

Translation Invariance

If a feature detector is useful in one location, it is useful everywhere.

The Parameter Miracle

A 3×3 convolution layer applying 32 filters to an RGB image requires only

864 parameters

Compared to our 150,000,000 parameter Dense layer, this is a ~99.999% reduction!

WHAT EXACTLY IS A KERNEL?



The Structure

Also known as a **Filter**. It is simply a very small, learnable weight matrix, typically 3×3 , 5×5 , or 7×7 .



The Purpose

It acts as a specific **Feature Detector**. During training, one kernel might learn to detect horizontal lines, another learns colors.



The Output

As it slides, it produces high numerical activations when it finds its matched pattern, and near-zero when it doesn't.

THE CONVOLUTION OPERATION

Image Patch × Kernel

I11	I12	I13
I21	I22	I23
I31	I32	I33

$$\text{Output} = \sum_{i=1}^k \sum_{j=1}^k I_{i,j} \cdot K_{i,j}$$

1. Element-wise Multiplication

Multiply overlapping pixels and weights.

2. Summation

Add all resulting products together.

3. Output

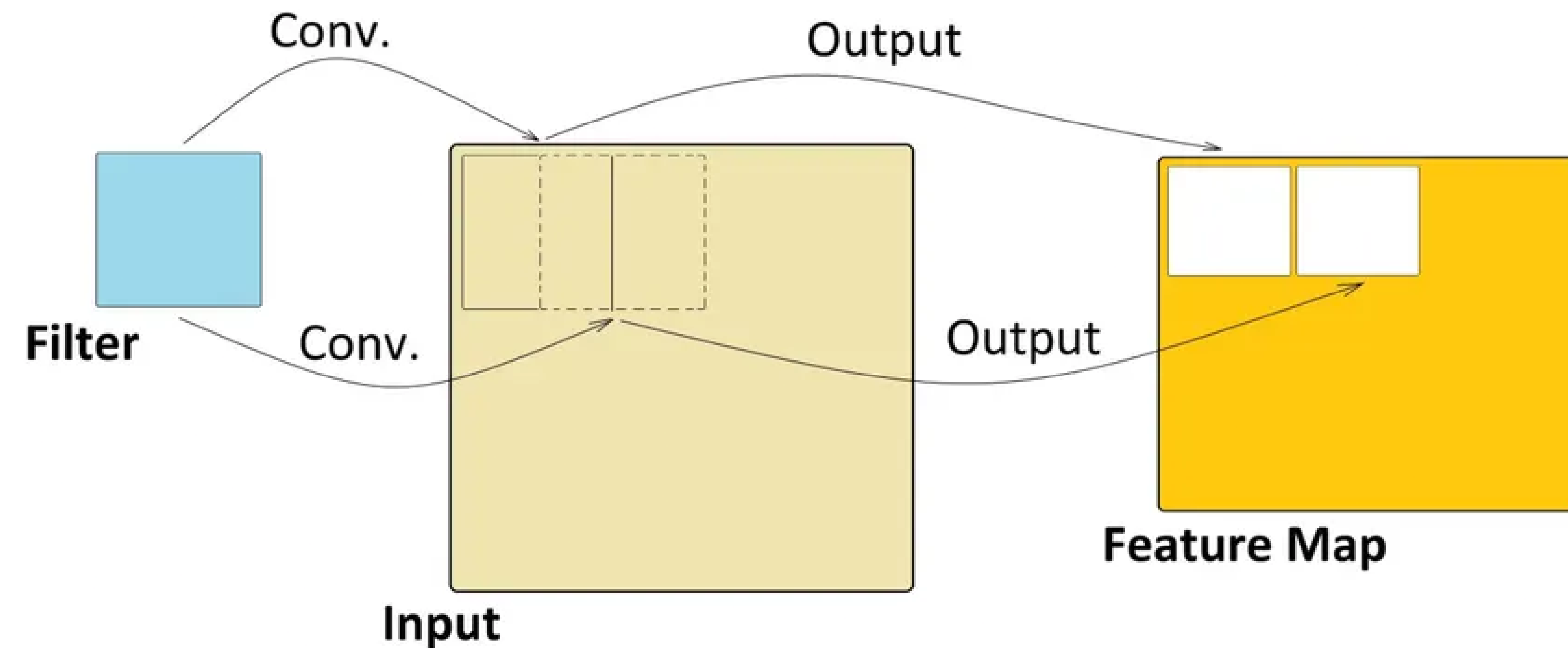
Place that single sum in the output matrix.

FEATURE MAPS (ACTIVATIONS)

When you slide a kernel entirely across an image, the resulting 2D output is called a **Feature Map**.

- It highlights exactly *where* the kernel found its specific pattern.
- A network learns many kernels per layer (e.g., 64).
- Therefore, one image input becomes 64 unique Feature Maps.

Think of it as looking at the world through different specialized lenses.



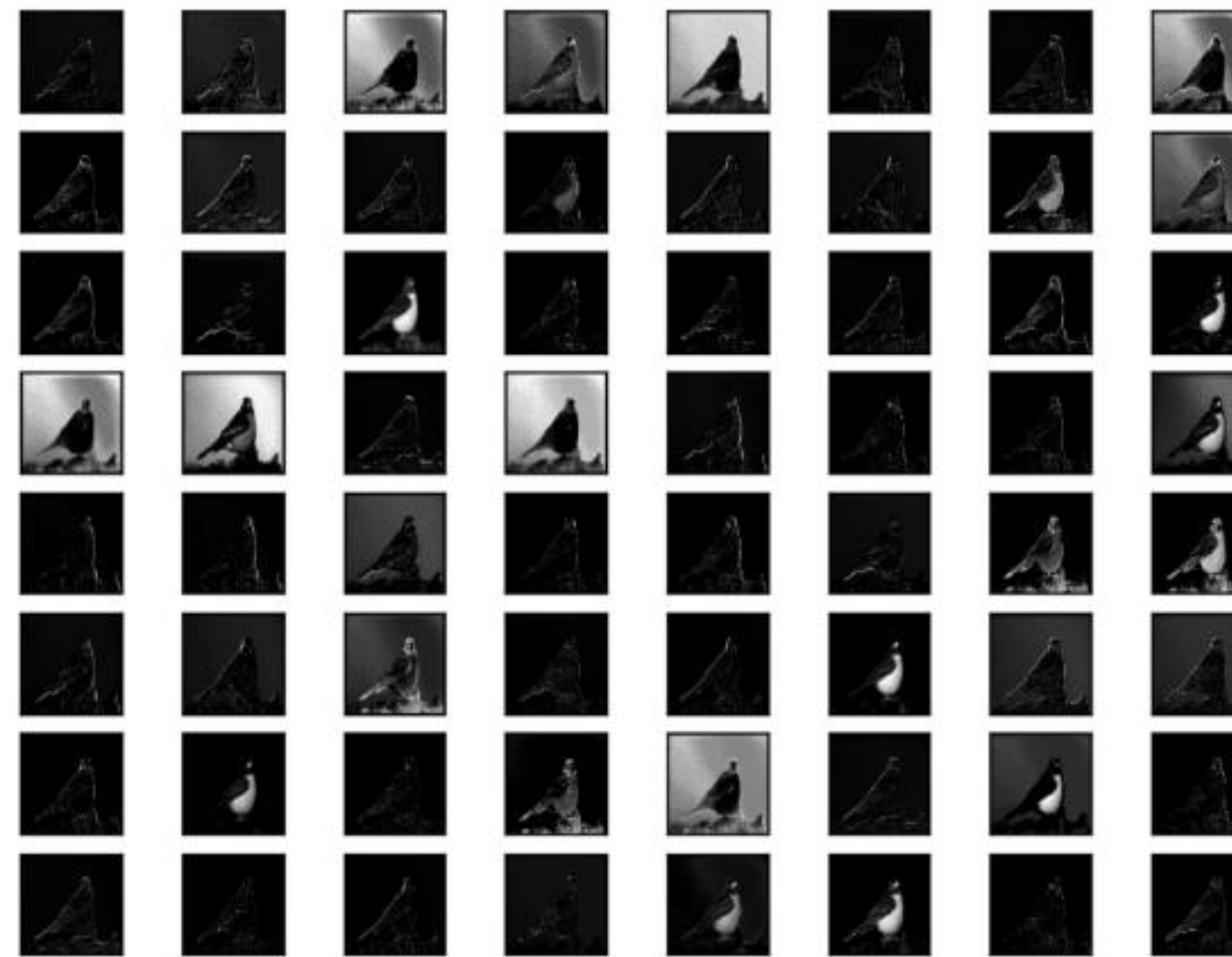
THE HIERARCHY OF FEATURES

By stacking convolution layers, the network learns a compositional hierarchy, piecing together simple shapes into complex semantics.



1. Low-Level

Edges, colors, gradients.



2. Mid-Level

Textures, corners, shapes.



3. High-Level

Object parts, faces, cars.

ARCHITECTURE COMPARISON

Attribute	Fully Connected (Dense)	Convolutional (CNN)
Connectivity	Global (All to All)	Local (Small neighborhoods)
Parameter Count	Astronomically High	Highly Efficient (Shared)
Spatial Structure	Destroyed via flattening	Preserved entirely
Translation	Vulnerable to shifts	Naturally Invariant
Suitability	Tabular Data, Final classifications	Images, Video, Audio spectrograms

CV APPLICATIONS

Because Convolutional Neural Networks perfectly solve the parameter explosion and spatial locality problems, they dominate modern Computer Vision.

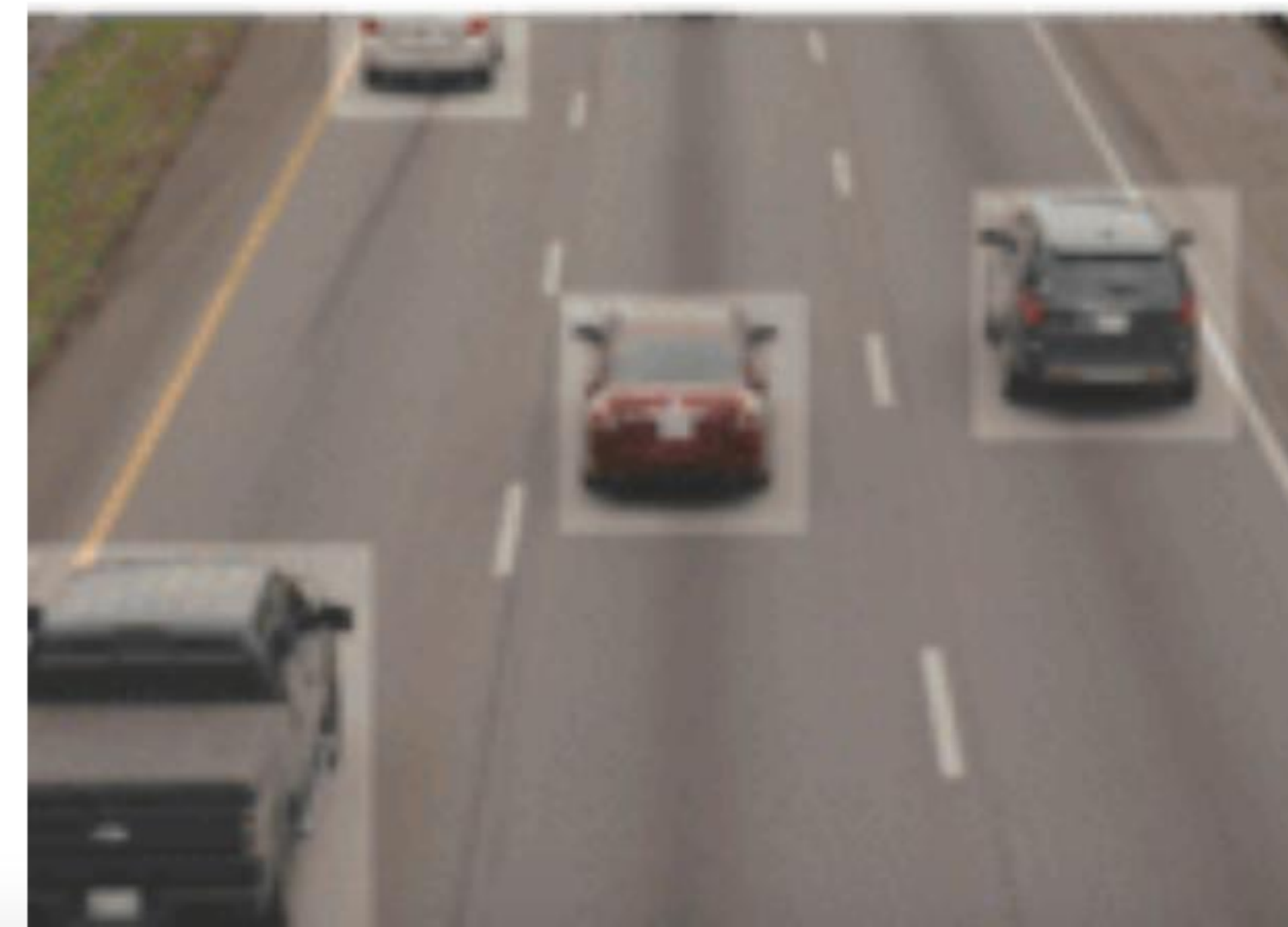
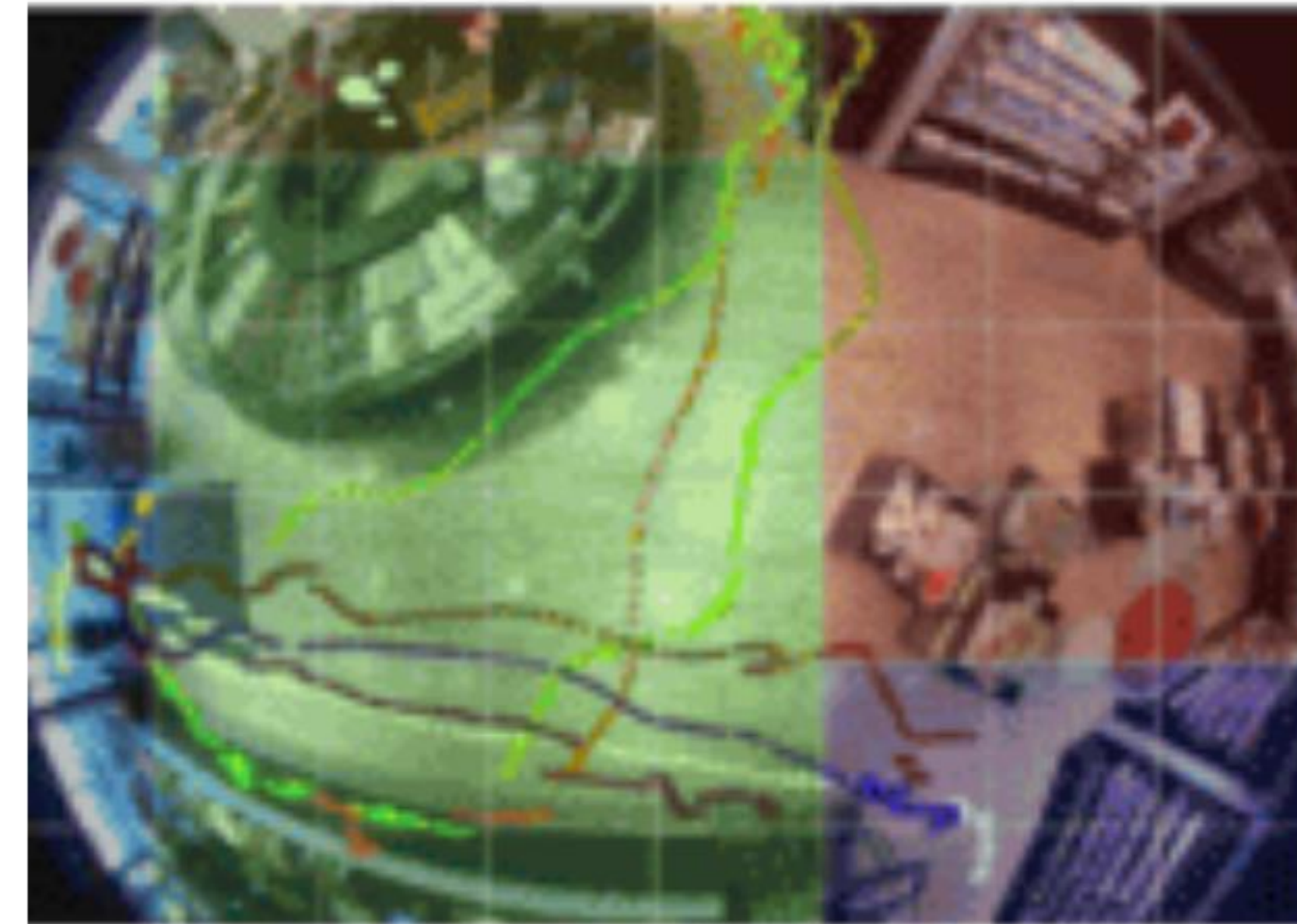
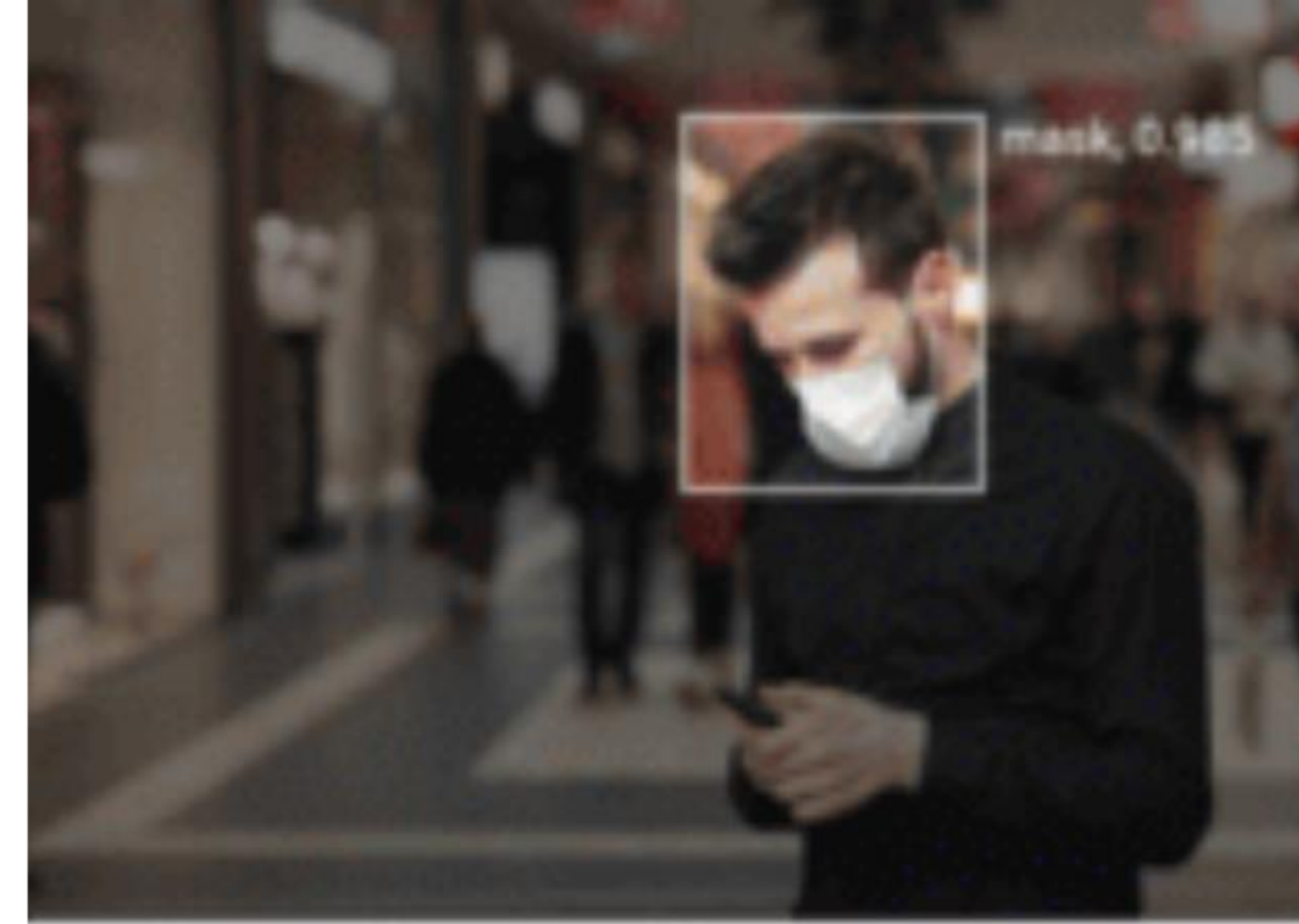
 **Autonomous Driving**

 **Medical Imaging (Tumor Det.)**

Facial Recognition

 **Satellite Crop Analysis**

 **Instructor Notes**



LECTURE SUMMARY

- ✓ **Spatial Structure is Key:** Images rely on 2D relationships. Flattening them into 1D vectors for dense networks destroys vital context.
- ✓ **Parameter Explosion:** Fully connected layers require impossible amounts of memory for high-resolution images.
- ✓ **Sliding Windows:** Convolutions extract features using small kernels that slide across the image, preserving geometry.
- ✓ **Efficiency via Sharing:** A single kernel's weights are reused across the entire image, drastically reducing parameters and achieving translation invariance.

Think-Pair-Share

"Why is using one small filter repeatedly more efficient than assigning a different weight to every pixel?"

Homework Challenge

Calculate the exact number of parameters required to process a $32 \times 32 \times 3$ image using:

1. A Fully Connected Layer with 1000 output neurons.
2. A single 3×3 Convolutional Filter.

Predict: Which architecture scales better to 4K resolution?

IMAGE SOURCES



<https://machinelearningmastery.com/wp-content/uploads/2019/02/Visualization-of-the-Feature-Maps-Extracted-from-the-Block-1-in-the-VGG16-Model-1024x768.png>

Source: machinelearningmastery.com



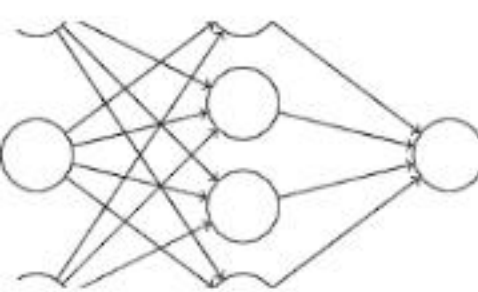
<https://media.easy-peasy.ai/82ec3354-d849-4d20-8b8c-dd73d29bf050/16bc59fa-de8e-46db-a14a-e7397a6d3ee2.png>

Source: easy-peasy.ai



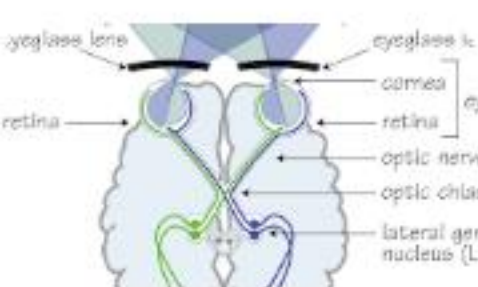
https://lh6.googleusercontent.com/_LTzvJ78g5GKI0zZFP5g1jZTy8isLbEjYgHiDgc1OvBvdX45zvxxlQizG5nRZtPmeU_dtK3ScbpQLAJ1rwGgkfIFDAHfyr1P9DsXbNiKVxgvhrVOexi1gllPnt9wiidocWgLqWLB

Source: www.topcoder.com



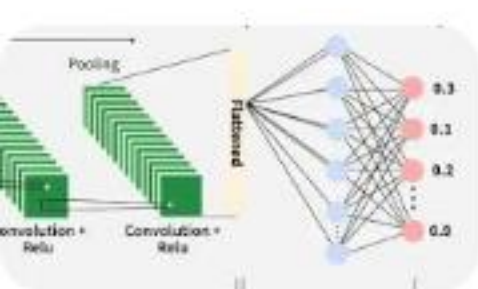
<https://deeplearningmath.org/images/tikz10.png>

Source: deeplearningmath.org



https://opticianworks.com/wp-content/uploads/2018/01/Human_visual_pathway-2-1.png

Source: opticianworks.com



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Source: www.geeksforgeeks.org

IMAGE SOURCES



https://images.stockcake.com/public/2/f/b/2fb793ee-9f6c-4508-9c3a-87e3fada80a5_large/connected-technology-mesh-stockcake.jpg

Source: stockcake.com



<https://viso.ai/wp-content/uploads/2021/12/computer-vision-applications-viso-ai-1060x370.png>

Source: viso.ai